Capacity, unit costs and a supermarket contract

Student tasks | Original TBS lesson for AQA A-level Business 7132 (exam cohorts of 2027, not the new 7138) | Section 3.4 Operational management

Crumb Lane Bakery has spare oven capacity. A supermarket offers a price below the bakery's current unit cost.

Before you start you should know: capacity utilisation = output / capacity x 100; unit cost = total cost / output; labour productivity = output / number of workers. You need a calculator. This is original classroom practice, not an exam paper. All business details and figures are fictional.

The bakery this week

Weekly position (fictional)	Amount
Capacity on one shift	2,000 loaves
Output and sales to regular customers	1,500 loaves
Regular selling price	£3.60 per loaf
Variable cost (flour, energy, packaging)	£0.80 per loaf
Fixed costs (rent, ovens, insurance, six salaried bakers)	£3,000
Bakers	6

Round money to the nearest penny, other figures to one decimal place. Ignore tax. Assume all loaves made are sold, bakers are salaried (a fixed cost, no overtime), no waste, and ovens can reach 100% of capacity.

Starter: which measure?

Match each to a measure: capacity utilisation, unit cost or labour productivity.

Description	Measure
(i) loaves made per baker	
(ii) average cost of one loaf	
(iii) share of oven capacity in use	

Task 1: the bakery today

a. Calculate capacity utilisation.

b. Calculate total cost and unit cost (average cost) per loaf.

c. Calculate labour productivity in loaves per baker.

d. Calculate weekly profit.

The contract offer

A supermarket offers a weekly contract for 800 loaves at £2.50 per loaf. This is lower than the bakery's unit cost in Task 1. The contract loaves cost the same £0.80 each to make. The bakery can still sell its 1,500 regular loaves at £3.60.

Task 2: why not simply say no?

a. Calculate the extra revenue, extra variable cost and extra profit from one contract loaf.

b. The owner says, "£2.50 is less than our £2.80 unit cost, so we would lose money." Explain whether the owner is right. Use your answer to part a.

Task 3: two ways to respond

Treat the options separately. Both start from the bakery in Task 1.

Option A: the supermarket accepts 500 loaves at £2.50 instead. They fit the spare capacity; fixed costs stay at £3,000 and the six bakers can make them.

Option B: accept all 800 contract loaves. The bakery adds a second shift that costs £1,000 a week in extra fixed costs (including the pay of the two extra bakers and other costs of running a second shift), raises capacity to 2,800 loaves and brings the team to 8 bakers.

Complete the table	Today	Option A	Option B
Total output (loaves)	1,500		
Capacity (loaves)	2,000		
Capacity utilisation (%)	75.0		
Fixed costs (£)	3,000		
Total cost (£)	4,200		
Unit cost (£ per loaf)	2.80		
Labour productivity (loaves per baker)	250.0		
Weekly profit (£)	1,200		

Working space for Options A and B:

Task 4: test the risk

The supermarket's own-label loaf is cheaper than the bakery's own. Some regular customers might switch to it.

a. Under Option A, 100 regular loaves a week are lost to the supermarket loaf. Calculate the new weekly profit. Assume the variable cost of the 100 loaves not made is saved.

b. Under Option A, what is the largest whole number of regular loaves the bakery could lose each week while the contract still gives more profit than declining (£1,200)? Hint: under Option A, each regular loaf lost reduces weekly profit by its contribution (price minus variable cost). Show your working.

Task 5: decide

Write a recommendation of about 80 words: accept A, accept B or decline. Use at least three figures. Explain how well each option uses capacity and name one risk or difficulty that could change your decision.

Exit ticket

a. Why can unit cost fall when output rises, even though variable cost per loaf stays the same?

b. Without the contract, the bakery makes 1,800 loaves instead of 1,500 on the same shift. Calculate its capacity utilisation and unit cost.